

STEFAN AMEYAW MILLS

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SUMMARY

Information Systems Ph.D. researcher and data scientist (4.0 GPA) with six years of applied experience across the United Nations and industry. I build and deploy machine learning, deep learning, and NLP models on large-scale, real-world data, from fraud-detection systems that protected over US\$1M to AI-generated-content detection at scale, and I pair them with statistics and econometrics to turn messy data into decisions. Strong across the full pipeline, from data engineering and modeling (Python, SQL, PyTorch) to communicating results clearly to non-technical stakeholders.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, transformers (BERT, RoBERTa), deep learning, NLP, classification & clustering, model evaluation, uncertainty quantification (Monte Carlo dropout), SHAP

Data & Programming: Python, R, SQL, ETL pipelines, large-scale data, feature engineering; Tableau, Power BI; AWS, Azure, Git

Quantitative: statistics, probability, applied econometrics, causal inference (difference-in-differences), event studies, simulation

EXPERIENCE

The University of Texas Rio Grande Valley Graduate Research Assistant Edinburg, TX | Sep 2025 - Present

- Build and fine-tune deep learning and NLP models (transformers: BERT, RoBERTa) on datasets of millions of records to detect AI-generated content.
- Quantify model uncertainty with Monte Carlo dropout, run large-scale data pipelines, and evaluate models with precision, recall, and F1.
- Apply econometrics and simulation to study trust and reliability in AI; co-author multiple research manuscripts.

United Nations World Food Programme Data Scientist Accra, Ghana | Apr 2024 - Aug 2025

- Built machine learning fraud-detection models in Python that prevented over US\$1M in donor losses.
- Architected ETL pipelines cutting manual workload 60%; engineered features and built predictive analytics on large-scale operational data.
- Delivered forecasting and trend analysis on food-security data via Tableau and Power BI dashboards used for humanitarian decisions.

Huawei Technologies Product Manager Accra, Ghana | Dec 2021 - Nov 2023

- Built ML anomaly-detection and forecasting models on large-scale 2G/3G/4G network data, improving efficiency and profitability by over 40%.
- Led a cross-functional team and data-driven market analytics, correlating performance with competitor and pricing trends to gain 30% market share in six months.
- Designed energy-optimization solutions that opened a new market segment, driving \$74K in revenue and \$800K in new business through vendor partnerships.

Huawei Technologies Wireless Engineer & Technical Team Lead Accra, Ghana | Sep 2019 - Dec 2021

- Analyzed large-scale network performance data to identify and troubleshoot bottlenecks, cutting downtime 95% across 800+ sites.
- Built automated data-processing and commissioning scripts in Python that improved workflow efficiency.
- Configured, integrated, and tested 4G/3G network elements across 800+ sites and mentored junior engineers, improving team proficiency 60%.

SELECTED PROJECTS

- **ADRD Clinical Decision Support (Trust-Calibrated AI in Healthcare).** Built an AI decision-support tool for early Alzheimer/dementia (ADRD) risk assessment with SHAP explanations, confidence scores, and AI recommendations; models clinician trust and reliance via Bayesian multi-agent simulation. (Python, ML, SHAP)
- **Synthetic Noise - AI-Generated Content Detection.** Ensemble of AI-text detectors (fine-tuned RoBERTa, DetectGPT, GPTZero) with Monte Carlo dropout for uncertainty, applied to millions of financial social-media records; difference-in-differences design.
- **CD&A Obfuscation and Market Reactions (SEC Item 402(x)).** NLP analysis of corporate compensation-disclosure text to measure obfuscation; event study linking obfuscation to market reactions (cumulative abnormal returns).

EDUCATION

Ph.D., Information Systems, The University of Texas Rio Grande Valley; GPA 4.0/4.0 Aug 2025 - Present

B.Sc., Computer Science, Kwame Nkrumah University of Science and Technology Jun 2019